

Aaweg Bhaladhare

GenAI Engineer · Multimodal RAG & Agent Infrastructure · LLM Evaluation & RLHF

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PROFILE

AI/ML engineer with **team-lead experience on LLM evaluation** at Turing (**Terminal Bench v2.0, OS Bench, SWE-Bench**). Ship production agentic systems, multimodal RAG, RLHF/SFT pipelines, and low-level inference infrastructure. Strongest in the gap between research-grade prototypes and shipped services.

PROFESSIONAL EXPERIENCE

Turing Enterprises Inc.

01/2025 – 04/2026

Team Lead — LLM Evaluation Engineer (rated “Top Performer”)

Remote

Clients: ByteDance · Alibaba · Tencent AI Lab · Meta AI · Microsoft · Apple

- Served as **Pod Lead** for **Terminal Bench** workflows — owned task distribution, review cycles, escalation handling, and quality alignment across contributor teams; rated “Top Performer.”
- Designed benchmark tasks evaluating frontier AI agents (Opus 4–6, GPT-5.x) on real terminal operations: debugging systems, process management, CI/CD troubleshooting, causal reasoning, and multi-domain shell scripting.
- Built validation scripts and automation pipelines verifying benchmark task correctness, test quality, and instruction–test alignment — reducing dataset error rates and enforcing isolation rules, metadata schemas, and submission standards.
- Contributed to **OSWorld** Phase 1 reworks and served as QA / Pod Lead for Phase 2 — refining interactive Linux desktop and application evaluation tasks, addressing non-deterministic UI behavior, pagination inconsistencies, and render-time layout constraints.
- Authored **SFT datasets** for math and code reasoning (Tencent AI Lab) and built tooling to convert Terminal Bench solutions into structured SFT notebooks in assistant / tool-call / tool-output format, with YAML-driven pipelines for reproducible dataset creation, validation, and reporting.
- Implemented **hallucination detection** and structure-alignment metrics; designed and validated **SWE-Bench** tasks assessing AI agents on real GitHub issues (bug fixes, feature implementations, refactoring) and built similarity-detection pipelines preventing duplicate benchmark tasks across contributor teams.
- Evaluated and benchmarked prompt-engineering strategies for **Meta’s LLMs** — instruction-following quality, reasoning depth, output alignment; authored structured rubrics and failure-pattern analysis identifying prompt-optimization opportunities across 22 task domains.
- Contributed to evaluation of Meta’s **OpenClaw** robotics manipulation framework — designed test scenarios and validation criteria measuring dexterous grasping precision, task completion rates, and generalization across manipulation environments.
- Directed quality rework of 50 client-flagged tasks — average score improvement **5.44 → 7.14 (+1.70)** across 21 approved tasks, with team review-score average **4.96 / 5.0**.
- Built a **Python automated LLM review pipeline** (OpenAI API, Anthropic API, Google Drive API) evaluating benchmark tasks across 22 domains with a 3-phase workflow: 9-dimension quality checks, client-calibrated scoring (qq/tq/tc), and a false-positive scanner; parallel batch processing with auto-downloaded task ZIPs and CSV / Excel / Markdown reports.
- Developed **Google Apps Script dashboards** (Rework 50 Task Tracker, Flagged 50 Status Tracker) providing real-time status, team contribution summaries, score improvement analytics, and priority-based action items across 50+ tasks.
- Ran knowledge-transfer sessions training engineers on benchmark design methodology, scoring standards, and evaluation consistency; served as primary escalation point for spec interpretation.

Tars Technologies

06/2024 – 01/2025

ML Engineer (Intern)

Pune, India

- Built a **YOLOv8** object-detection model for mine-worker safety monitoring, reaching **95% accuracy** on protective-gear detection.
- Shipped a face-recognition attendance system automating tracking for **50+ employees**, cutting manual error by ~30%.
- Contributed to a full-stack LMS (Next.js 13, React, Prisma, Tailwind, MySQL) with focus on the model-serving and inference-API layer.
- Mentored junior engineers on containerization, model integration patterns, and code-review hygiene.

FEATURED PROJECTS

FastAPI · LangGraph · LangChain · LlamaIndex · ChromaDB · Gemini 3.x · CLIP ViT-B/32 · Groq Whisper · MLflow · Airflow · Docker · GCP Cloud Run · Hugging Face Spaces

[github.com/Aawegg/mosaicmind-rag](#) · [huggingface.co/spaces/Aawegg/mosaicmind](#)

- End-to-end agentic **multimodal RAG** over PDF, image, audio, and video on a **FastAPI** service with **SSE token streaming**, multipart upload, eval, and corpus-management routes; single-page chat UI with multi-conversation history.
- Agent designed as a **LangGraph** state machine — *plan* → *retrieve* → *synthesize* → *reflect* — with a self-critique loop that re-queries when citations are insufficient.
- Hybrid retrieval over two **ChromaDB** collections in **LlamaIndex**: Gemini text embeddings + **CLIP (CNN vision encoder)** image embeddings, fused at the prompt layer with deterministic citation chips.
- Full MLOps loop: **MLflow** tracking on every query (latency, citations, plan), **Airflow** DAGs for nightly re-index + weekly eval sweeps, **pytest + ruff** in **GitHub Actions CI**, multi-stage **Docker**.
- Shipped to **GCP Cloud Run** (Workload Identity Federation, Artifact Registry, Secret Manager) and **Hugging Face Spaces** (GitOps auto-mirror), both gated behind GitHub Actions.

Python · Pydantic · google-genai SDK · scipy · GitHub Actions CI · Docker

[github.com/Aawegg/priorityjudge](#)

- Built a multi-pass agent (extractor → 5 per-criterion scorers → deterministic citation verifier → synthesizer) on **Gemini 3.x** for plan-quality auditing.
- Scored **9887/10000** on a hand-rated calibration set with **perfect Spearman** rank correlation against the human grader, **96.5% citation precision**, and **99.6% test-retest stability**.
- **28 passing tests** covering rubric integrity, citer determinism, prompt-render regression, and a no-secrets repo scan.

C++ · ARM NEON SIMD · Docker-in-Docker · Python · llama.cpp

[github.com/Aawegg/nibblecore](#)

- Architected an LLM evaluation framework that runs **OS-level benchmarks** inside secure, automated **Docker-in-Docker sandboxes**.
- Wrote **low-level C++ quantization kernels** using **ARM NEON SIMD intrinsics**, beating standard **llama.cpp** dequantization throughput on Apple Silicon.
- Designed recursive multi-agent evaluators that autonomously score LLM reasoning and instruction-following inside dynamic system environments.
- Built a model-routing layer that cascades inference between fast-response and deep-reasoning models to optimize latency and API cost.

OPEN-SOURCE CONTRIBUTIONS

[PR #21625](#) · [Issue #21610](#)

- Fixed file-descriptor and HTTP-connection leaks in the **HotpotQA** evaluation benchmark — wrapped the chunked-download path and the dataset reader in proper **context managers** (Python 3.10+ parenthesized with syntax) and switched `json.loads(f.read())` to `json.load(f)`.
- Diagnosed the leak from a fresh read of `llama-index-core/llama_index/core/evaluation/benchmarks/hotpotqa.py`, opened the issue with a reproduction plan, then shipped the patch against the same plan.

SKILLS

Programming	Python · C / C++ · TypeScript · SQL · Bash
LLMs & Agents	Gemini 3.x · Llama-3.x · LangChain · LangGraph · LlamaIndex · RAG · Multi-agent orchestration · Function calling · Prompt engineering
ML / DL	PyTorch · Hugging Face Transformers · SentenceTransformers · CLIP (CNN) · YOLOv8 · OpenCV · Fine-tuning (LoRA / QLoRA) · Quantization (GGUF) · NLP
Eval & RLHF	SFT / RLHF trajectory authoring · Terminal Bench v2.0 · OS Bench · SWE-Bench · LLM-as-judge · automated retrieval & faithfulness scoring · hallucination detection · Spearman / Kendall calibration · rubric design
Backend / APIs	FastAPI · Pydantic · Server-Sent Events · REST · async Python · uvicorn
MLOps / Infra	Docker · docker-compose · MLflow · Airflow · GitHub Actions CI/CD · pytest · ruff · pre-commit
Cloud	GCP (Cloud Run, Artifact Registry, Secret Manager, Workload Identity Federation) · Hugging Face Spaces · AWS (familiar) · Oracle Cloud

Vector / Data

Soft skills

ChromaDB · embeddings (Gemini, CLIP, SentenceTransformers) · PyMuPDF · ffmpeg · OpenCV
Team leadership · Mentoring · Quick adaptability · Cross-functional communication · Problem-solving

EDUCATION

B.Tech, Artificial Intelligence & Data Science — Vishwakarma Institute of Information Technology, Pune

2021 – 04/2025 · CGPA 8.54

CERTIFICATIONS

- IBM Data Science Specialization
 - IBM Machine Learning
 - Microsoft Power BI Data Analyst
 - Career Essentials in Generative AI — Microsoft & LinkedIn
 - Databases and SQL for Data Science with Python